

Filter Summary Report: CG,TIA,Automated,simple,Z2,Z3

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated simple **Z2 Z3**: $\frac{Z_2Z_3g_m+Z_3}{Z_2g_m+1}$

$$H(z) = \frac{Z_2Z_3g_m + Z_3}{Z_2g_m + 1}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(\infty, \ R_2, \ \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3}, \ \infty, \ \infty, \ \infty \right)$

$$H(s) = \frac{L_3R_3s}{C_3L_3R_3s^2 + L_3s + R_3}$$

Parameters:

Q: $\frac{\sqrt{C_3}R_3}{\sqrt{L_3}}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{1}{C_3R_3}$
K-LP: 0
K-HP: 0
K-BP: R_3
Qz: None
Wz: None

3.2 BP-2 $Z(s) = \left(\infty, \ \frac{1}{C_2s}, \ \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3}, \ \infty, \ \infty, \ \infty \right)$

$$H(s) = \frac{L_3R_3s}{C_3L_3R_3s^2 + L_3s + R_3}$$

Parameters:

Q: $\frac{\sqrt{C_3}R_3}{\sqrt{L_3}}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{1}{C_3R_3}$
K-LP: 0
K-HP: 0
K-BP: R_3
Qz: None
Wz: None

3.3 BP-3 $Z(s) = \left(\infty, \ \frac{R_2}{C_2R_2s+1}, \ \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3}, \ \infty, \ \infty, \ \infty \right)$

$$H(s) = \frac{L_3R_3s}{C_3L_3R_3s^2 + L_3s + R_3}$$

Parameters:

Q: $\frac{\sqrt{C_3}R_3}{\sqrt{L_3}}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{1}{C_3R_3}$
K-LP: 0
K-HP: 0
K-BP: R_3
Qz: None
Wz: None

3.4 BP-4 $Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_3} R_3}{\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
 bandwidth: $\frac{1}{C_3 R_3}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

$$H(s) = \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}$$

3.5 BP-5 $Z(s) = \left(\infty, L_2 s + \frac{1}{C_2 s}, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_3} R_3}{\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
 bandwidth: $\frac{1}{C_3 R_3}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

$$H(s) = \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}$$

3.6 BP-6 $Z(s) = \left(\infty, L_2 s + R_2 + \frac{1}{C_2 s}, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_3} R_3}{\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
 bandwidth: $\frac{1}{C_3 R_3}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

$$H(s) = \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}$$

3.7 BP-7 $Z(s) = \left(\infty, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \infty \right)$

Parameters:

Q: $\frac{\sqrt{C_3} R_3}{\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
 bandwidth: $\frac{1}{C_3 R_3}$
 K-LP: 0
 K-HP: 0

$$H(s) = \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}$$

K-BP: R_3
 Qz: None
 Wz: None

3.8 BP-8 $Z(s) = \left(\infty, \frac{R_2(C_2L_2s^2+1)}{C_2L_2s^2+C_2R_2s+1}, \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3}, \infty, \infty, \infty \right)$

$$H(s) = \frac{L_3R_3s}{C_3L_3R_3s^2 + L_3s + R_3}$$

Parameters:

Q: $\frac{\sqrt{C_3R_3}}{\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
 bandwidth: $\frac{1}{C_3R_3}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(\infty, R_2, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_3L_3R_3s^2 + R_3}{C_3L_3s^2 + C_3R_3s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_3}}{\sqrt{C_3R_3}}$
 wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
 bandwidth: $\frac{R_3}{L_3}$
 K-LP: R_3
 K-HP: R_3
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

4.2 BS-2 $Z(s) = \left(\infty, \frac{1}{C_2s}, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_3L_3R_3s^2 + R_3}{C_3L_3s^2 + C_3R_3s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_3}}{\sqrt{C_3R_3}}$
 wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
 bandwidth: $\frac{R_3}{L_3}$
 K-LP: R_3
 K-HP: R_3
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

$$4.3 \quad \text{BS-3} \quad Z(s) = \left(\infty, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

Parameters:

$$\begin{aligned} \text{Q:} & \frac{\sqrt{L_3}}{\sqrt{C_3 R_3}} \\ \text{wo:} & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \\ \text{bandwidth:} & \frac{R_3}{L_3} \\ \text{K-LP:} & R_3 \\ \text{K-HP:} & R_3 \\ \text{K-BP:} & 0 \\ \text{Qz:} & \text{None} \\ \text{Wz:} & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \end{aligned}$$

$$4.4 \quad \text{BS-4} \quad Z(s) = \left(\infty, \quad R_2 + \frac{1}{C_2 s}, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

Parameters:

$$\begin{aligned} \text{Q:} & \frac{\sqrt{L_3}}{\sqrt{C_3 R_3}} \\ \text{wo:} & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \\ \text{bandwidth:} & \frac{R_3}{L_3} \\ \text{K-LP:} & R_3 \\ \text{K-HP:} & R_3 \\ \text{K-BP:} & 0 \\ \text{Qz:} & \text{None} \\ \text{Wz:} & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \end{aligned}$$

$$4.5 \quad \text{BS-5} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

Parameters:

$$\begin{aligned} \text{Q:} & \frac{\sqrt{L_3}}{\sqrt{C_3 R_3}} \\ \text{wo:} & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \\ \text{bandwidth:} & \frac{R_3}{L_3} \\ \text{K-LP:} & R_3 \\ \text{K-HP:} & R_3 \\ \text{K-BP:} & 0 \\ \text{Qz:} & \text{None} \\ \text{Wz:} & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \end{aligned}$$

$$4.6 \quad \text{BS-6} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

Parameters:

$$\begin{aligned} \text{Q:} & \frac{\sqrt{L_3}}{\sqrt{C_3 R_3}} \\ \text{wo:} & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \\ \text{bandwidth:} & \frac{R_3}{L_3} \end{aligned}$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + R_3}{C_3 L_3 s^2 + C_3 R_3 s + 1}$$

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K-LP: R_3
K-HP: R_3
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

4.7 **BS-7** $Z(s) = \left(\infty, \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1}, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_3L_3R_3s^2 + R_3}{C_3L_3s^2 + C_3R_3s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_3}}{\sqrt{C_3}R_3}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{R_3}{L_3}$
K-LP: R_3
K-HP: R_3
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

4.8 **BS-8** $Z(s) = \left(\infty, \frac{R_2(C_2L_2s^2+1)}{C_2L_2s^2+C_2R_2s+1}, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_3L_3R_3s^2 + R_3}{C_3L_3s^2 + C_3R_3s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_3}}{\sqrt{C_3}R_3}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{R_3}{L_3}$
K-LP: R_3
K-HP: R_3
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

5 **GE**

6 **HP**

7 **LP**

8 **X-INVALID-NUMER**

9 **X-INVALID-ORDER**

$$\mathbf{9.1 \quad X-INVALID-ORDER-1} \quad Z(s) = (\infty, \ R_2, \ R_3, \ \infty, \ \infty, \ \infty)$$

$$H(s) = R_3$$

$$\mathbf{9.2 \quad X-INVALID-ORDER-2} \quad Z(s) = \left(\infty, \ R_2, \ \frac{1}{C_3 s}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.3 \quad X-INVALID-ORDER-3} \quad Z(s) = \left(\infty, \ R_2, \ \frac{R_3}{C_3 R_3 s + 1}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.4 \quad X-INVALID-ORDER-4} \quad Z(s) = \left(\infty, \ R_2, \ R_3 + \frac{1}{C_3 s}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{R_2 g_m + s (C_3 R_2 R_3 g_m + C_3 R_3) + 1}{s (C_3 R_2 g_m + C_3)}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(\infty, \ R_2, \ L_3 s + \frac{1}{C_3 s}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^2 (C_3 L_3 R_2 g_m + C_3 L_3) + 1}{s (C_3 R_2 g_m + C_3)}$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(\infty, \ R_2, \ \frac{L_3 s}{C_3 L_3 s^2 + 1}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{L_3 s}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(\infty, \ R_2, \ L_3 s + R_3 + \frac{1}{C_3 s}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^2 (C_3 L_3 R_2 g_m + C_3 L_3) + s (C_3 R_2 R_3 g_m + C_3 R_3) + 1}{s (C_3 R_2 g_m + C_3)}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(\infty, \ R_2, \ \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\infty, \ \frac{1}{C_2 s}, \ R_3, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = R_3$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \ \frac{1}{C_2 s}, \ \frac{1}{C_3 s}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_3 R_3 s^2 + g_m + s(C_2 + C_3 R_3 g_m)}{C_2 C_3 s^2 + C_3 g_m s}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_3 s^3 + C_2 s + C_3 L_3 g_m s^2 + g_m}{C_2 C_3 s^2 + C_3 g_m s}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_3 s}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_3 s^3 + g_m + s^2(C_2 C_3 R_3 + C_3 L_3 g_m) + s(C_2 + C_3 R_3 g_m)}{C_2 C_3 s^2 + C_3 g_m s}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, R_3, \infty, \infty, \infty \right)$$

$$H(s) = R_3$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_3 R_2 R_3 s^2 + R_2 g_m + s(C_2 R_2 + C_3 R_2 R_3 g_m + C_3 R_3) + 1}{C_2 C_3 R_2 s^2 + s(C_3 R_2 g_m + C_3)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_3 R_2 s^3 + C_2 R_2 s + R_2 g_m + s^2 (C_3 L_3 R_2 g_m + C_3 L_3) + 1}{C_2 C_3 R_2 s^2 + s (C_3 R_2 g_m + C_3)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_3 s}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_3 R_2 s^3 + R_2 g_m + s^2 (C_2 C_3 R_2 R_3 + C_3 L_3 R_2 g_m + C_3 L_3) + s (C_2 R_2 + C_3 R_2 R_3 g_m + C_3 R_3) + 1}{C_2 C_3 R_2 s^2 + s (C_3 R_2 g_m + C_3)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, R_3, \infty, \infty, \infty \right)$$

$$H(s) = R_3$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_2 C_3 R_2 R_3 g_m + C_2 C_3 R_3) + s (C_2 R_2 g_m + C_2 + C_3 R_3 g_m)}{C_3 g_m s + s^2 (C_2 C_3 R_2 g_m + C_2 C_3)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 g_m s^2 + g_m + s^3 (C_2 C_3 L_3 R_2 g_m + C_2 C_3 L_3) + s (C_2 R_2 g_m + C_2)}{C_3 g_m s + s^2 (C_2 C_3 R_2 g_m + C_2 C_3)}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_3 s}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(\infty, \quad R_2 + \frac{1}{C_2 s}, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_2 C_3 L_3 R_2 g_m + C_2 C_3 L_3) + s^2 (C_2 C_3 R_2 R_3 g_m + C_2 C_3 R_3 + C_3 L_3 g_m) + s (C_2 R_2 g_m + C_2 + C_3 R_3 g_m)}{C_3 g_m s + s^2 (C_2 C_3 R_2 g_m + C_2 C_3)}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(\infty, \quad R_2 + \frac{1}{C_2 s}, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad R_3, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = R_3$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad R_3 + \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_2 R_3 g_m s^3 + g_m + s^2 (C_2 C_3 R_3 + C_2 L_2 g_m) + s (C_2 + C_3 R_3 g_m)}{C_2 C_3 L_2 g_m s^3 + C_2 C_3 s^2 + C_3 g_m s}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad L_3 s + \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_2 L_3 g_m s^4 + C_2 C_3 L_3 s^3 + C_2 s + g_m + s^2 (C_2 L_2 g_m + C_3 L_3 g_m)}{C_2 C_3 L_2 g_m s^3 + C_2 C_3 s^2 + C_3 g_m s}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 s}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_2 L_3 g_m s^4 + g_m + s^3 (C_2 C_3 L_2 R_3 g_m + C_2 C_3 L_3) + s^2 (C_2 C_3 R_3 + C_2 L_2 g_m + C_3 L_3 g_m) + s (C_2 + C_3 R_3 g_m)}{C_2 C_3 L_2 g_m s^3 + C_2 C_3 s^2 + C_3 g_m s}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(\infty, \quad L_2 s + \frac{1}{C_2 s}, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.41 \quad X-INVALID-ORDER-41} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad R_3, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = R_3$$

$$\mathbf{9.42 \quad X-INVALID-ORDER-42} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.43 \quad X-INVALID-ORDER-43} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad \frac{R_3}{C_3 R_3 s + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.44 \quad X-INVALID-ORDER-44} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad R_3 + \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_2 R_3 g_m s^3 + g_m + s^2 (C_2 C_3 R_2 R_3 g_m + C_2 C_3 R_3 + C_2 L_2 g_m) + s (C_2 R_2 g_m + C_2 + C_3 R_3 g_m)}{C_2 C_3 L_2 g_m s^3 + C_3 g_m s + s^2 (C_2 C_3 R_2 g_m + C_2 C_3)}$$

$$\mathbf{9.45 \quad X-INVALID-ORDER-45} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad L_3 s + \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_2 L_3 g_m s^4 + g_m + s^3 (C_2 C_3 L_3 R_2 g_m + C_2 C_3 L_3) + s^2 (C_2 L_2 g_m + C_3 L_3 g_m) + s (C_2 R_2 g_m + C_2)}{C_2 C_3 L_2 g_m s^3 + C_3 g_m s + s^2 (C_2 C_3 R_2 g_m + C_2 C_3)}$$

$$\mathbf{9.46 \quad X-INVALID-ORDER-46} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad \frac{L_3 s}{C_3 L_3 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{L_3 s}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.47 \quad X-INVALID-ORDER-47} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad L_3 s + R_3 + \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_3 L_2 L_3 g_m s^4 + g_m + s^3 (C_2 C_3 L_2 R_3 g_m + C_2 C_3 L_3 R_2 g_m + C_2 C_3 L_3) + s^2 (C_2 C_3 R_2 R_3 g_m + C_2 C_3 R_3 + C_2 L_2 g_m + C_3 L_3 g_m) + s (C_2 R_2 g_m + C_2 + C_3 R_3 g_m)}{C_2 C_3 L_2 g_m s^3 + C_3 g_m s + s^2 (C_2 C_3 R_2 g_m + C_2 C_3)}$$

$$\mathbf{9.48 \quad X-INVALID-ORDER-48} \quad Z(s) = \left(\infty, \quad L_2 s + R_2 + \frac{1}{C_2 s}, \quad \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}$$

$$\mathbf{9.49 \quad X-INVALID-ORDER-49} \quad Z(s) = \left(\infty, \quad \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \quad R_3, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = R_3$$

$$\mathbf{9.50 \quad X-INVALID-ORDER-50} \quad Z(s) = \left(\infty, \quad \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \quad \frac{1}{C_3 s}, \quad \infty, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$9.51 \quad \mathbf{X-INVALID-ORDER-51} \quad Z(s) = \left(\infty, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$9.52 \quad \mathbf{X-INVALID-ORDER-52} \quad Z(s) = \left(\infty, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^3 (C_2 C_3 L_2 R_2 R_3 g_m + C_2 C_3 L_2 R_3) + s^2 (C_2 L_2 R_2 g_m + C_2 L_2 + C_3 L_2 R_3 g_m) + s (C_3 R_2 R_3 g_m + C_3 R_3 + L_2 g_m) + 1}{C_3 L_2 g_m s^2 + s^3 (C_2 C_3 L_2 R_2 g_m + C_2 C_3 L_2) + s (C_3 R_2 g_m + C_3)}$$

$$9.53 \quad \mathbf{X-INVALID-ORDER-53} \quad Z(s) = \left(\infty, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_2 L_3 g_m s^3 + L_2 g_m s + R_2 g_m + s^4 (C_2 C_3 L_2 L_3 R_2 g_m + C_2 C_3 L_2 L_3) + s^2 (C_2 L_2 R_2 g_m + C_2 L_2 + C_3 L_3 R_2 g_m + C_3 L_3) + 1}{C_3 L_2 g_m s^2 + s^3 (C_2 C_3 L_2 R_2 g_m + C_2 C_3 L_2) + s (C_3 R_2 g_m + C_3)}$$

$$9.54 \quad \mathbf{X-INVALID-ORDER-54} \quad Z(s) = \left(\infty, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_3 s}{C_3 L_3 s^2 + 1}$$

$$9.55 \quad \mathbf{X-INVALID-ORDER-55} \quad Z(s) = \left(\infty, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^4 (C_2 C_3 L_2 L_3 R_2 g_m + C_2 C_3 L_2 L_3) + s^3 (C_2 C_3 L_2 R_2 R_3 g_m + C_2 C_3 L_2 R_3 + C_3 L_2 L_3 g_m) + s^2 (C_2 L_2 R_2 g_m + C_2 L_2 + C_3 L_2 R_3 g_m + C_3 L_3 R_2 g_m + C_3 L_3) + s (C_3 R_2 R_3 g_m + C_3 R_3 + L_2 g_m) + 1}{C_3 L_2 g_m s^2 + s^3 (C_2 C_3 L_2 R_2 g_m + C_2 C_3 L_2) + s (C_3 R_2 g_m + C_3)}$$

$$9.56 \quad \mathbf{X-INVALID-ORDER-56} \quad Z(s) = \left(\infty, \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1}, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}$$

$$9.57 \quad \mathbf{X-INVALID-ORDER-57} \quad Z(s) = \left(\infty, \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, R_3, \infty, \infty, \infty \right)$$

$$H(s) = R_3$$

$$9.58 \quad \mathbf{X-INVALID-ORDER-58} \quad Z(s) = \left(\infty, \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$9.59 \quad \mathbf{X-INVALID-ORDER-59} \quad Z(s) = \left(\infty, \frac{R_2 (C_2 L_2 s^2 + 1)}{C_2 L_2 s^2 + C_2 R_2 s + 1}, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(\infty, \frac{R_2(C_2L_2s^2+1)}{C_2L_2s^2+C_2R_2s+1}, R_3 + \frac{1}{C_3s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2g_m + s^3(C_2C_3L_2R_2R_3g_m + C_2C_3L_2R_3) + s^2(C_2C_3R_2R_3 + C_2L_2R_2g_m + C_2L_2) + s(C_2R_2 + C_3R_2R_3g_m + C_3R_3) + 1}{C_2C_3R_2s^2 + s^3(C_2C_3L_2R_2g_m + C_2C_3L_2) + s(C_3R_2g_m + C_3)}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(\infty, \frac{R_2(C_2L_2s^2+1)}{C_2L_2s^2+C_2R_2s+1}, L_3s + \frac{1}{C_3s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2C_3L_3R_2s^3 + C_2R_2s + R_2g_m + s^4(C_2C_3L_2L_3R_2g_m + C_2C_3L_2L_3) + s^2(C_2L_2R_2g_m + C_2L_2 + C_3L_3R_2g_m + C_3L_3) + 1}{C_2C_3R_2s^2 + s^3(C_2C_3L_2R_2g_m + C_2C_3L_2) + s(C_3R_2g_m + C_3)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(\infty, \frac{R_2(C_2L_2s^2+1)}{C_2L_2s^2+C_2R_2s+1}, \frac{L_3s}{C_3L_3s^2+1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{L_3s}{C_3L_3s^2 + 1}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(\infty, \frac{R_2(C_2L_2s^2+1)}{C_2L_2s^2+C_2R_2s+1}, L_3s + R_3 + \frac{1}{C_3s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2g_m + s^4(C_2C_3L_2L_3R_2g_m + C_2C_3L_2L_3) + s^3(C_2C_3L_2R_2R_3g_m + C_2C_3L_2R_3 + C_2C_3L_3R_2) + s^2(C_2C_3R_2R_3 + C_2L_2R_2g_m + C_2L_2 + C_3L_3R_2g_m + C_3L_3) + s(C_2R_2 + C_3R_2R_3g_m + C_3R_3) + 1}{C_2C_3R_2s^2 + s^3(C_2C_3L_2R_2g_m + C_2C_3L_2) + s(C_3R_2g_m + C_3)}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(\infty, \frac{R_2(C_2L_2s^2+1)}{C_2L_2s^2+C_2R_2s+1}, \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_3L_3R_3s^2 + L_3s + R_3}{C_3L_3s^2 + 1}$$

10 X-INVALID-WZ

11 X-PolynomialError